

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

Answer:

1. Understanding of the Problem and Constraints

- **Remote Rural Location:** The branch office is situated in a geographically isolated area lacking high-speed terrestrial wired broadband (such as fiber optic or copper DSL cables).
- **Critical Application Demands:** The network must support real-time video conferencing (requiring low latency and minimal jitter), cloud-based SaaS applications (e.g., Microsoft 365, ERP/CRM), and secure encrypted traffic exchange with corporate headquarters (HQ).
- **Strict Budget Constraints:** Limited budget for WAN connectivity rules out expensive dedicated private circuits (e.g., MPLS leased lines).
- **Uninterrupted Operations (High Availability):** Business continuity requires zero single-points-of-failure (SPOF) and seamless automated failover mechanisms.
- **Engineering Assumptions Note:** Traffic volume is sized for a typical branch of 15–25 concurrent employees; final bandwidth profiles and link budgets should be validated via a physical site survey and link feasibility test.

2. Problem Decomposition

To design an operational and resilient remote network within budget constraints, the architecture is decomposed into five distinct functional sub-systems:

1. **WAN Edge & Physical Connectivity:** Selecting high-speed wireless WAN technologies to compensate for the absence of wired infrastructure.

- 2. SD-WAN & Dynamic Path Control:** Deploying Software-Defined WAN (SD-WAN) for real-time traffic monitoring, load balancing, and automated failover.
- 3. LAN Infrastructure & PoE Switching:** Managed switching and Wi-Fi 6 wireless coverage for local network clients.
- 4. Security & Encrypted Tunnelling:** Next-Generation Firewall (NGFW) with Site-to-Site IPsec VPN, 802.1X Network Access Control, and multi-factor authentication.
- 5. QoS & Traffic Optimization:** Quality of Service (QoS) queueing and bandwidth shaping to prioritize latency-sensitive applications over background traffic.

3. Proposed Network Design

- **Primary WAN Connection:** Low-Earth-Orbit (LEO) Satellite Internet (e.g., Starlink Business). Delivers 100–220 Mbps download and 10–25 Mbps upload with low latency (~25–40 ms), making it highly suitable for real-time video conferencing and cloud applications compared to traditional high-latency (~600 ms) geostationary satellites.
- **Secondary/Backup WAN Connection:** Dual-SIM 4G LTE / 5G Cellular Gateway operating on two separate Mobile Network Operators (MNOs). Acts as an active-standby or load-balanced secondary circuit for continuous connectivity.
- **Networking Devices Selection:** Recommended devices include:
 - **Edge Router / SD-WAN Gateway:** Enterprise SD-WAN Router (e.g., Fortinet FortiGate 60F or Cisco Meraki MX67) equipped with multi-WAN interface support, dynamic path selection, and integrated VPN acceleration.
 - **Managed Access Switch:** 24-Port Gigabit Managed PoE+ Switch (IEEE 802.3at) providing 802.1Q VLAN isolation, 802.1X port-level security, and inline power for access points and IP phones.
 - **Wireless Access Point:** Enterprise Wi-Fi 6 (802.11ax) dual-band indoor AP supporting WPA3-Enterprise, multi-SSID mapping, and seamless roaming.
 - **Firewall / Security Gateway:** Integrated NGFW module on the SD-WAN appliance providing Stateful Packet Inspection (SPI), Intrusion Prevention System (IPS), Web Filtering, and Antivirus protection.
- **Logical Network Topology:** Star / Hub-and-Spoke topology. The rural branch acts as a spoke node establishing dual encrypted IPsec VPN tunnels back to the corporate HQ Hub gateway.

4. Application of Constraints in Solution

- **Bandwidth Limitations:** Solved by implementing DSCP-based Quality of Service (QoS). Real-time video streams (DSCP EF / AF41) and critical cloud SaaS traffic are prioritized in hardware queues, while non-essential web traffic and background updates are rate-limited.
- **Uninterrupted Operations (Reliability):** Automated SD-WAN sub-second failover. If the LEO satellite experiences brief rain fade or obstruction, active sessions switch automatically to the 5G/4G cellular backup link without dropping VPN or voice calls.
- **Budget Control:** Replaces multi-thousand-dollar monthly MPLS leases with cost-effective commercial LEO satellite and cellular data plans, dramatically lowering OpEx while maintaining high bandwidth.
- **Future Scalability:** SD-WAN architecture simplifies future expansion; additional WAN links (e.g., future rural fiber) can be plugged directly into the edge appliance without changing internal IP schemes or network topology.

5. Security and Reliability Mechanisms

- **Site-to-Site Encrypted Tunnels:** IPsec VPN with AES-256-GCM encryption, SHA-384 data integrity, and Diffie-Hellman Group 14/21 for Perfect Forward Secrecy (PFS).
- **Authentication & Access Control:** 802.1X Network Access Control (NAC) tied to HQ Active Directory / RADIUS server. Wi-Fi users authenticate via WPA3-Enterprise; remote management requires Multi-Factor Authentication (MFA).

- **Threat Defense:** Zero-Trust Network Access (ZTNA) policies, IPS signature scanning, and automated threat prevention at the network edge.

6. Cost, Performance, and Scalability Optimization

- **Performance Optimization:** Local Internet Breakout (Direct Internet Access - DIA) routes public SaaS traffic (Microsoft 365, Google Workspace) directly to the cloud via the LEO satellite, while only internal corporate traffic is backhauled over the IPsec tunnel to HQ, minimizing tunnel overhead.
- **CapEx/OpEx Efficiency:** Standardizing on all-in-one desktop NGFW/SD-WAN edge units reduces upfront hardware investment and ongoing maintenance expenses.

7. Logical Justification, Innovation, and Conclusion

- **Innovative Approach:** Combining Low-Earth-Orbit satellite technology with Dual-MNO cellular failover under an intelligent SD-WAN overlay turns two non-traditional rural mediums into an enterprise-grade high-availability connection.
- **Conclusion:** The recommended SD-WAN + LEO/Cellular hybrid network directly solves all remote rural constraints. It delivers high bandwidth for video/cloud applications, robust IPsec/MFA security for corporate communication, sub-second failover for uninterrupted operations, and strict adherence to limited budget boundaries.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer:

1. Understanding of the Problem and Constraints

- **High Simultaneous User Density:** Serves a campus community of 3,000+ students, with an estimated 6,000–9,000 active wireless client devices (smartphones, laptops, tablets) operating simultaneously across classrooms, labs, libraries, and hostels.
- **Diverse Environmental Requirements:** Needs to cover varied spaces—high-density static learning zones (lecture halls, libraries), physical obstacle zones (hostel walls), and high RF noise areas (laboratories).
- **Budget & Access Point Constraints:** Restricted AP quantity requires optimizing AP placement and channel utilization for maximum capacity per unit rather than blanket coverage over-deployment.
- **Interference Control:** High user density risks severe Co-Channel Interference (CCI) and Adjacent Channel Interference (ACI) if radio spectrum is not meticulously managed.
- **Engineering Assumptions Note:** Capacity models assume ~60–70% simultaneous active client concurrency (~1,800–2,100 concurrent devices); final AP quantities and antenna patterns must be confirmed via a predictive and active onsite RF site survey (e.g., Ekahau AI Pro).

2. Problem Decomposition

To achieve an optimal balance across all constraints, the campus Wi-Fi network is decomposed into four strategic domains:

1. **Capacity-Based RF & AP Placement:** Allocating access points based on client density and throughput demands rather than geographical distance alone.

2. **Spectrum & Channel Engineering:** Structuring 2.4 GHz, 5 GHz (and 6 GHz) frequency bands, transmit powers, and channel widths to eliminate CCI.
3. **Centralized Architecture & PoE Switching:** Implementing Wireless LAN Controllers (WLC) and Layer 3 PoE+ switches with VLAN segmentation.
4. **Enterprise Security & Access Policy:** Deploying 802.1X / RADIUS, WPA3-Enterprise, guest portals, and airtime fairness.

3. Proposed Network Design

- **AP Placement & Density Strategy:** Strategic deployment per zone:
 - **Classrooms, Lecture Halls & Libraries (High Density):** High-capacity Wi-Fi 6 (802.11ax) 4x4 MU-MIMO Access Points ceiling-mounted overhead. AP transmit power is intentionally lowered (to 10–14 dBm) to shrink cell sizes, forcing client distribution across multiple APs and preventing cell overload.
 - **Hostels & Dormitories (Wall Obstructions):** Staggered wall-plate Wi-Fi 6 APs placed in corridors and study areas to penetrate concrete/brick partitions while servicing student rooms.
 - **Laboratories & Outdoor Common Areas:** Targeted APs equipped with directional sector antennas to focus RF coverage and mitigate laboratory equipment interference.
- **Frequency Band Selection & Channel Planning:** Dual-band management:
 - **5 GHz Frequency Band (Primary Data Carrier):** Prioritized for all student laptops, smartphones, and video/academic applications. Configured with 20 MHz or 40 MHz channel widths to maximize non-overlapping channels (UNII-1, UNII-2, UNII-3) and avoid co-channel overlap.
 - **2.4 GHz Frequency Band (Legacy Carrier):** Strictly limited to non-overlapping Channels 1, 6, and 11. Transmit power is reduced by 6–9 dB relative to 5 GHz (or disabled on every second AP) to mitigate severe 2.4 GHz propagation overlap.
 - **Band Steering & Airtime Fairness:** Band steering actively directs dual-band clients to 5 GHz. Airtime fairness prevents slow legacy devices (802.11g/n) from monopolizing channel time.
- **Network Devices & Infrastructure:** Centralized control elements:
 - **Centralized Wireless LAN Controller (WLC):** Redundant Hardware/Cloud WLC running Dynamic Channel Assignment (DCA), Transmit Power Control (TPC), and fast roaming (802.11k/v/r).
 - **Core & Distribution Switches:** Layer 3 Managed PoE+ (802.3at) Switches with 10GbE uplinks to guarantee switch backhaul capacity.
- **Authentication & Network Segmentation:** Role-based network access:
 - **Student & Staff Access:** WPA2/WPA3-Enterprise authentication using 802.1X EAP-PEAP/TTLS integrated with university Active Directory / LDAP.
 - **Guest Network:** WPA3-Personal / Open with Captive Portal, SMS/Email OTP registration, strict client isolation, and rate-limiting (e.g., max 5 Mbps per guest).
 - **VLAN Segmentation:** VLAN 10 (Faculty/Staff), VLAN 20 (Classrooms/Labs), VLAN 30 (Hostels), VLAN 40 (Guests), VLAN 99 (AP Management).

4. Application of Constraints in Solution

- **Limited AP Count vs High Density:** Solved by capacity-based cell sizing, band steering to 5 GHz, and Wi-Fi 6 OFDMA / MU-MIMO technology which allows a single AP to serve dozens of concurrent streams efficiently.
- **Signal Interference Minimization:** Achieved via non-overlapping channel maps, dynamic auto-RF adjustment (Radio Resource Management - RRM), and controlled transmit power.

- **Budget Efficiency:** Concentrates high-spec APs in high-density student aggregation zones (libraries, lecture halls) while using cost-effective placement in low-density administrative areas.

5. Security and Reliability Mechanisms

- **Mutual Authentication:** 802.1X EAP authentication prevents rogue access point spoofing and eavesdropping.
- **Wireless Intrusion Prevention System (WIPS):** Monitors radio spectrum for rogue APs, MAC spoofing, and wireless deauthentication attacks.
- **Role-Based Access Control (RBAC):** Layer 3 ACLs on core switches enforce isolation between student VLANs and administrative database servers.

6. Cost, Performance, and Scalability Optimization

- **Performance:** Wi-Fi 6 (802.11ax) multi-user scheduling eliminates packet collisions in crowded lecture halls.
- **Cost Control:** Utilizing existing campus fiber backhaul and adopting virtualized/cloud WLC software reduces hardware capital expenditure.
- **Scalability:** The centralized WLC framework allows adding new APs seamlessly as new university blocks are constructed.

7. Logical Justification: Balancing Coverage, Performance, Security, and Cost

- **Balancing Matrix:** Evaluation across four core metrics:
 - **Coverage:** Delivered through strategic, obstacle-aware AP positioning without blind spots.
 - **Performance:** Maximized via 5 GHz band steering, 802.11ax MU-MIMO, and reduced cell overlapping.
 - **Security:** Guaranteed by 802.1X WPA3-Enterprise, RADIUS integration, and complete VLAN separation.
 - **Cost:** Optimized by capacity-targeted deployment, avoiding over-provisioning while using cloud management.
- **Conclusion:** The proposed Wi-Fi architecture delivers a high-performance, secure campus wireless environment for 3,000+ users while strictly operating within budget and resource constraints.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer:

1. Understanding of the Problem and Constraints

- **High-Security Financial Domain:** The institution processes sensitive customer financial records, online banking transactions, and Personally Identifiable Information (PII), making it a prime target for cyber threats.
- **Strict Compliance Mandates:** Must strictly comply with financial security standards (e.g., PCI-DSS, ISO/IEC 27001, local banking regulations).

- **Performance & Zero Latency Demands:** Security controls (firewalls, deep packet inspection, encryption) must operate at line rate without degrading online banking or internal transaction performance.
- **Operational Budget Constraints:** Must maximize cybersecurity effectiveness without ballooning hardware CapEx or administrative OpEx.
- **Engineering Assumptions Note:** Sizing assumes gigabit internal backbone routing and redundant active-passive firewall pairs; specific inspection throughput should be verified against peak transaction volume.

2. Problem Decomposition

The architecture is structured into five defensive layers following a zero-trust model:

1. **Network Segmentation & Security Zones:** Dividing the network into isolated VLANs (DMZ, Employee Workstations, Core Servers, Database/PCI Zone).
2. **Perimeter & Internal Firewall Architecture:** Deploying Next-Generation Firewalls (NGFW) at the edge and Internal Segmentation Firewalls (ISFW) core.
3. **Intrusion Detection & Centralized Monitoring:** Implementing inline Intrusion Prevention Systems (IPS) and Security Information and Event Management (SIEM).
4. **Identity, Authentication & Access Control:** Port security (802.1X), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC).
5. **High Availability & Compliance Controls:** Redundant active-passive hardware pairs, encrypted transit, and audit logging.

3. Proposed Network Design

- **Network Segmentation Strategy (VLANs & Security Zones):** Micro-segmentation architecture:
 - **Zone 1: Public DMZ (VLAN 100):** Hosts public web applications, online banking portals, and external API gateways. Completely isolated from internal user networks.
 - **Zone 2: Internal Employee Workstations (VLAN 200):** Teller and administrative staff PCs. Restricted to accessing business application servers over controlled ports.
 - **Zone 3: Core Application Server Zone (VLAN 300):** Middleware and business logic servers processing banking functions.
 - **Zone 4: High-Security Database / PCI-DSS Zone (VLAN 400):** Houses core databases and sensitive customer financial data. Strictly isolated; accessible only by authorized application servers over specific, encrypted database ports.
 - **Zone 5: Management & Security Zone (VLAN 500):** SIEM, Active Directory, RADIUS servers, and network management tools.
- **Firewall Deployment Strategy:** Two-tier firewall topology:
 - **Perimeter Next-Generation Firewall (NGFW):** Positioned at the internet edge to inspect incoming customer web traffic, protect against volumetric DDoS attacks, block malicious IPs, and terminate SSL/TLS VPN connections for remote staff.
 - **Internal Segmentation Firewall (ISFW):** Positioned between internal user VLANs, Application Servers, and the Database/PCI Zone. Enforces zero-trust internal traffic filtering using stateful inspection and deep packet inspection (DPI).
- **Intrusion Detection & Prevention System (IDS/IPS):** Inline IPS modules active on both NGFW and ISFW to inspect traffic for signature matches, protocol anomalies, SQL injection attempts, and unauthorized lateral movement.
- **Centralized Security Monitoring (SIEM):** Centralized SIEM engine collecting syslog events from firewalls, switches, domain controllers, and databases for real-time threat detection and automated security alerting.

4. Application of Constraints in Solution

- **Regulatory Compliance (PCI-DSS):** Fully satisfies Requirement 1 (firewall protection between public networks and cardholder data), Requirement 2 (removal of default settings), Requirement 7 (restricting access to cardholder data by business need-to-know), and Requirement 10 (tracking and auditing all network access to cardholder data).
- **Performance Preservation:** The ISFW utilizes hardware ASIC acceleration to perform stateful and deep packet inspection at wire-speed (10 Gbps+), preventing latency bottlenecks on core financial transactions.
- **Budget Control:** Reuses existing Layer 3 core switch fabric for VLAN routing while deploying targeted ISFW policies at critical zone boundaries, avoiding the exorbitant cost of deploying firewalls on every individual switch port.

5. Security and Reliability Mechanisms

- **Port-Level & User Authentication:** Identity controls:
 - **Port Security:** 802.1X EAP-TLS with digital machine certificates prevents unauthorized rogue laptops from connecting to office Ethernet drops.
 - **User Authentication:** Multi-Factor Authentication (MFA) mandatory for all user network logins, administrative sessions, and remote VPN connections.
 - **Access Policy:** Role-Based Access Control (RBAC) enforcing the Principle of Least Privilege (e.g., bank tellers cannot initiate direct connections to the core database).
- **Data Encryption Standards:** TLS 1.3 for all web services and internal server-to-database sessions; IPsec with AES-256 for site-to-site communication.
- **High Availability (HA):** Active-Passive NGFW/ISFW clustering with stateful connection failover ensures sub-second recovery in the event of a primary firewall hardware failure.

6. Cost, Performance, and Scalability Optimization

- **Performance:** Intra-VLAN traffic routes locally at line rate on Layer 3 switches, while inter-zone cross-talk undergoes hardware-accelerated ISFW inspection.
- **Cost Efficiency:** Consolidating security services (NGFW, IPS, VPN, Content Inspection) into unified threat management platforms reduces licensing overhead and operational complexity.

7. Logical Justification: Attack Surface Reduction & Lateral Movement Limitation

- **Why Segmentation Reduces Attack Surface & Limits Lateral Movement:** Security rationale:
 - **Flaws of Flat Networks:** In a flat, unsegmented network, an attacker compromising a teller workstation (VLAN 200) via phishing or malware immediately gains unrestricted network access to the core database containing customer records.
 - **Micro-segmentation Containment:** In this proposed micro-segmented architecture, the ISFW sits between VLAN 200 (Employees) and VLAN 400 (Database). The ISFW enforces explicit stateful rules that deny all direct workstation-to-database communication.
 - **Incident Isolation:** If a workstation is compromised, the breach is strictly contained within VLAN 200. The attacker cannot pivot laterally to the database or DMZ. The inline IPS detects the lateral scanning attempt, drops the packets, and alerts the SIEM SOC team instantly.
- **Conclusion:** The multi-zoned network design delivers a zero-trust, defense-in-depth security architecture that satisfies strict PCI-DSS/financial compliance rules, protects customer data, and maintains high performance within budget boundaries.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand